



PROGRAM STUDI  
TEKNOLOGI REKAYASA  
PERANGKAT LUNAK  
POLIBATAM



# Final PBL Report

Your Project Title

## PROJECT IDENTITY

|                                                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PBL ID Number</b>                                      | : | PBL-TRPL412                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Project Proposer</b>                                   | : |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>Project Title</b>                                      | : | Rancang Bangun Node Sensor Nirkabel Berbasis LoRa untuk Monitoring Anomali Suhu Ruang Server Menggunakan Analisis Data Berbasis AI                                                                                                                                                                                                                                                                                                                         |
| <b>Output</b>                                             | : | IoT Application, Final PBL Report, Poster, User Manual, Demo Video, Draft of Intellectual Property Document                                                                                                                                                                                                                                                                                                                                                |
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# CHAPTER 1: INTRODUCTION

## Objectives

This system aims to develop an Internet of Things (IoT)-based monitoring solution to track temperature and humidity in server rooms in real time using LoRa communication. With this system, the monitoring process becomes more efficient by reducing the need for continuous manual checking, as data is collected and transmitted automatically. The temperature and humidity data provided are accurate and can be accessed in real time through a monitoring dashboard, making it easier for users to continuously observe environmental conditions.

In addition, this system applies Artificial Intelligence (AI) techniques by utilizing both temperature and humidity data to detect anomalies in server room conditions in a more adaptive way. This approach allows the system not only to identify extreme conditions but also to recognize subtle changes that may lead to potential risks. The analysis results are then used to classify environmental conditions into three categories: normal, warning, and critical, based on data patterns identified by AI rather than relying solely on fixed threshold values.

Overall, this system is designed not just as a monitoring tool, but as a decision-support system that can provide early warnings of potential issues in server environments. Its implementation is expected to improve the reliability of server operations, minimize the risk of damage caused by unstable environmental conditions, and support more effective and proactive IT infrastructure management.

## Scope

The scope of this system development includes the design and implementation of a server room monitoring system that focuses on temperature and humidity parameters using Internet of Things (IoT) technology with LoRa communication. The system consists of three main components: a sensor node, a LoRa gateway, and a monitoring server.

The sensor node utilizes a TTGO LoRa32 device integrated with a DHT22 sensor to periodically collect temperature and humidity data. The collected data is transmitted to the gateway using low-power LoRa communication.

The gateway is responsible for receiving data from the sensor node and forwarding it to the server using the MQTT protocol over a WiFi connection. The monitoring server handles data storage in a database and performs data processing, including anomaly detection using Artificial Intelligence (AI) based on temperature and humidity inputs.

The system also includes the development of a monitoring dashboard that displays real-time data and the results of the environmental condition classification.

## Problem Statements

1. Temperature and humidity monitoring in server rooms is still performed manually, which increases the risk of delayed detection of abnormal environmental changes.
2. The absence of a monitoring system that provides real-time temperature and humidity data makes it difficult to identify potential overheating and system disturbances at an early stage.

3. Uncontrolled increases in temperature and humidity can reduce server performance and increase the risk of hardware damage.
4. There is no anomaly detection system based on Artificial Intelligence (AI) that can analyze patterns of temperature and humidity changes adaptively.
5. Limitations in efficient and low-power long-range communication systems pose challenges in implementing IoT-based monitoring in server environments.

## Benefits

1. Enables real-time monitoring of temperature and humidity in server rooms.
2. Helps detect potential anomalies at an early stage before they become critical.
3. Reduces the risk of server damage caused by unstable environmental conditions.
4. Supports more controlled and data-driven server room management.

## Document Overview

This final report is structured to provide a comprehensive overview of the design and implementation of an IoT-based server room monitoring system using LoRa communication and Artificial Intelligence (AI) for anomaly detection.

Chapter 1 presents the introduction, including background, problem statements, objectives, scope, and benefits of the proposed system.

Chapter 2 describes the general overview of the system, including product perspective, main features, system benefits, and user characteristics.

Chapter 3 explains the system development methods, covering IoT architecture design, communication mechanisms, and the AI-based approach used for anomaly detection.

Chapter 4 outlines the specific system requirements, including system overview, design scheme, and operational specifications.

Chapter 5 discusses the system implementation, which includes hardware implementation, software development, and database management.

Chapter 6 presents the system testing results and the conclusions of the project.

## CHAPTER 2: GENERAL DESCRIPTION

### Product Perspective

The proposed system is an IoT-based monitoring solution designed to observe environmental conditions in a server room, specifically temperature and humidity. It operates as a distributed system consisting of three main components: a sensor node, a LoRa gateway, and a monitoring server.

The sensor node, built using a TTGO LoRa32 device integrated with a DHT22 sensor, functions as the data acquisition unit that periodically reads temperature and humidity values. These data are transmitted wirelessly to the gateway using LoRa communication, which is suitable for low-power and long-range data transmission.

The gateway acts as an intermediary between the sensor node and the server. It receives data from the node via LoRa and forwards it to the monitoring server using the MQTT protocol over a WiFi connection.

The monitoring server serves as the central processing unit of the system. It is responsible for storing incoming data, performing data processing, and executing anomaly detection using Artificial Intelligence (AI) based on temperature and humidity parameters. The processed data and analysis results are then presented through a monitoring dashboard.

Within this perspective, the system is positioned as an integrated IoT solution that combines data acquisition, wireless communication, and intelligent data analysis to support environmental monitoring in server rooms.

### Product Features

Below are the main features that will be developed in this application:

1. The system is capable of periodically collecting temperature and humidity data using a DHT22 sensor integrated with a TTGO LoRa32 sensor node.
2. The collected data are transmitted wirelessly from the sensor node to the gateway using LoRa communication.
3. The gateway receives data from the sensor node and forwards it to the monitoring server using the MQTT protocol over a WiFi connection.
4. The monitoring server stores the received temperature and humidity data in a database for further processing.
5. The system performs data analysis using Artificial Intelligence (AI) by utilizing both temperature and humidity parameters to detect anomalies.
6. The system classifies environmental conditions into three categories: normal, warning, and critical based on the analysis results.
7. The system provides a monitoring dashboard to display real-time data and classification results.

## Product Benefits

The benefits obtained from using this system include:

1. Provides continuous monitoring of temperature and humidity in server rooms without relying on manual checking.
2. Enables early identification of abnormal environmental conditions through data-driven analysis.
3. Reduces the risk of server performance degradation and hardware damage caused by unstable temperature and humidity.
4. Supports more effective monitoring through centralized data storage and real-time visualization.
5. Improves operational awareness by providing clear environmental status classification based on analyzed data.

## User Characteristics

In this application, the involved users are:

1. The primary users are server administrators and IT support staff responsible for maintaining server room conditions.
2. Users have basic to intermediate technical knowledge in computer systems and networking.
3. Users are familiar with monitoring tools and are able to interpret temperature and humidity data.
4. Users require a simple and clear interface to observe environmental conditions and system status.
5. Users rely on the system for routine monitoring and early awareness of abnormal conditions in server environments.



## CHAPTER 3: BASIC METHODS OF PROJECT DEVELOPMENT

### Design Method

The development method used in this system follows a prototype-based approach combined with functional and structural design. The system is developed through iterative stages starting from requirement analysis, system design, implementation, and testing. The development process is carried out offline by the project team at the Software Engineering Technology Laboratory, Batam State Polytechnic. The system prototype is built and refined gradually based on system requirements related to IoT-based monitoring of temperature and humidity in server rooms. The development activities include designing the IoT architecture (sensor node, gateway, and server), implementing LoRa communication, integrating MQTT for data transmission, and developing an AI-based anomaly detection model. Regular evaluation and refinement are conducted to ensure the system aligns with functional requirements defined in the project specification. The overall development process is carried out within a defined project period of approximately four months.

### Technology Design

#### Functional Design

- **Database and Server**

The system utilizes a centralized server architecture responsible for managing data storage and processing. Incoming data from IoT devices is received through an MQTT communication channel and processed by a backend service developed using FastAPI. The processed data, including temperature and humidity readings, are stored in a MySQL database for further analysis and historical tracking. The server also handles API requests from the monitoring dashboard to provide real-time data visualization.

- **IoT Devices**

The IoT subsystem consists of a sensor node and a gateway based on TTGO LoRa32 devices. The sensor node is equipped with a DHT22 sensor to measure temperature and humidity periodically. The collected data is transmitted from the sensor node to the gateway using LoRa communication. The gateway receives the data and forwards it to the server via MQTT protocol over a WiFi connection. This layered architecture ensures reliable long-range communication and efficient data delivery to the central system.

- **Artificial Intelligence (AI)**

The system implements an Artificial Intelligence (AI) module to analyze environmental data based on temperature and humidity parameters. Data preprocessing is performed to normalize the input values before analysis. Anomaly detection is conducted using statistical methods such as Z-score and machine learning techniques such as Isolation Forest to identify abnormal patterns. The AI module generates an anomaly score, which is then used to classify environmental conditions into three categories: normal, warning, and critical.

## Electrical Design

- **Mechanical Design.** Mechanical design of the system consists of a compact placement of the TTGO LoRa32 module and DHT22 sensor inside a protective casing. The casing is designed to ensure proper airflow for accurate temperature and humidity measurement while protecting the components from physical damage and environmental interference in the server room.
- **Electrical Design.** The electrical design consists of a TTGO LoRa32 microcontroller integrated with a DHT22 sensor as the main data acquisition unit. The system is powered by a stable power supply and uses LoRa communication for wireless data transmission between the sensor node and gateway. The gateway forwards data to the monitoring server via MQTT over WiFi for further processing and analysis.

## Communication Design

The communication design of the system is based on a multi-layer architecture consisting of a sensor node, a gateway, and a monitoring server. The sensor node transmits temperature and humidity data collected from the DHT22 sensor to the gateway using LoRa communication, which is suitable for low-power and long-range transmission. The gateway acts as an intermediary device that receives data from the sensor node and forwards it to the monitoring server using the MQTT protocol over a WiFi connection. This design ensures reliable data transfer from the field layer to the server layer. At the server level, incoming data is processed, stored in a database, and analyzed using Artificial Intelligence (AI) to detect anomalies in environmental conditions based on temperature and humidity parameters.

## The stages of IoT device development

The production process begins with determining and procuring all types of materials and tools to be used. This aims to optimize the manufacturing process. The production process includes:

1. **Requirement Analysis** The system requirements are identified based on the need for real-time monitoring of temperature and humidity in server rooms, including data acquisition, transmission, and analysis using IoT and AI technologies.
2. **System Design** The overall architecture is designed consisting of three main components: sensor node, LoRa gateway, and monitoring server. The communication flow between devices is defined using LoRa and MQTT protocols.
3. **Hardware Implementation** The sensor node is implemented using a TTGO LoRa32 device integrated with a DHT22 sensor to collect temperature and humidity data. A gateway device is also configured to receive LoRa data and forward it to the server.
4. **Communication Setup** LoRa communication is configured between the sensor node and gateway, while MQTT over WiFi is used to transmit data from the gateway to the server.
5. **Software Development** The server-side system is developed to handle data storage, processing, and visualization. Anomaly detection is implemented using Artificial Intelligence (AI) based on temperature and humidity parameters.
6. **System Testing** The system is tested to ensure proper data transmission from node to server, as well as correct detection and classification of environmental conditions into normal, warning, and critical states.

By following these steps, the production process can proceed efficiently, leading to the creation of a high-quality tool that meets all specifications and requirements

## Product Testing Stage

There are several stages in product testing, namely:

### 1. Hardware Testing

This stage focuses on testing the functionality of the sensor node and gateway, including the TTGO LoRa32 device and DHT22 sensor, to ensure that temperature and humidity data can be read correctly.

### 2. Communication Testing

This stage evaluates the data transmission process using LoRa communication between the sensor node and the gateway, as well as MQTT communication between the gateway and the server.

### 3. System Integration Testing

This stage tests the integration of all system components, including data flow from the sensor node to the monitoring server and database storage.

### 4. Data Processing and AI Testing

This stage evaluates the system's ability to process temperature and humidity data and perform anomaly detection using Artificial Intelligence (AI) methods.

### 5. Dashboard Testing

This stage verifies the functionality of the monitoring dashboard in displaying real-time data and classification results.

## CHAPTER 4: SPECIFIC REQUIREMENT

### System Overview

The proposed system is an IoT-based monitoring system designed to observe temperature and humidity conditions in server rooms in real time. The system is built using a distributed architecture consisting of three main components: a sensor node, a LoRa gateway, and a monitoring server. The sensor node, implemented using a TTGO LoRa32 device integrated with a DHT22 sensor, is responsible for collecting temperature and humidity data periodically from the environment. This data is then transmitted wirelessly to the gateway using LoRa communication. The gateway acts as an intermediary device that receives data from the sensor node and forwards it to the monitoring server using the MQTT protocol over a WiFi network. This ensures reliable data transmission between the edge device and the central system. The monitoring server functions as the central processing unit of the system. It handles data storage, data management, and processing of incoming environmental data. The server also performs anomaly detection based on temperature and humidity parameters using Artificial Intelligence (AI) techniques. The processed data and analysis results are then made available through a monitoring dashboard, allowing users to observe real-time environmental conditions and system status.

### Design and Equipment Scheme

The system design consists of a distributed IoT architecture that includes a sensor node, a LoRa gateway, and a monitoring server. Each component is designed to work together to enable real-time environmental monitoring in a server room. The sensor node is built using a TTGO LoRa32 device integrated with a DHT22 sensor. This node is responsible for collecting temperature and humidity data from the environment at regular intervals. The collected data is transmitted from the sensor node to the gateway using LoRa communication technology, which is selected for its low power consumption and long-range capability. The gateway device, also based on TTGO LoRa32, receives the data from the sensor node and forwards it to the monitoring server using the MQTT protocol over a WiFi connection. The monitoring server consists of a software-based system running on a computer or server environment. It handles data reception, storage in a MySQL database, and processing using Artificial Intelligence (AI) for anomaly detection based on temperature and humidity parameters. The overall system design ensures continuous data flow from the sensor node to the server, enabling real-time monitoring and analysis of environmental conditions in the server room.

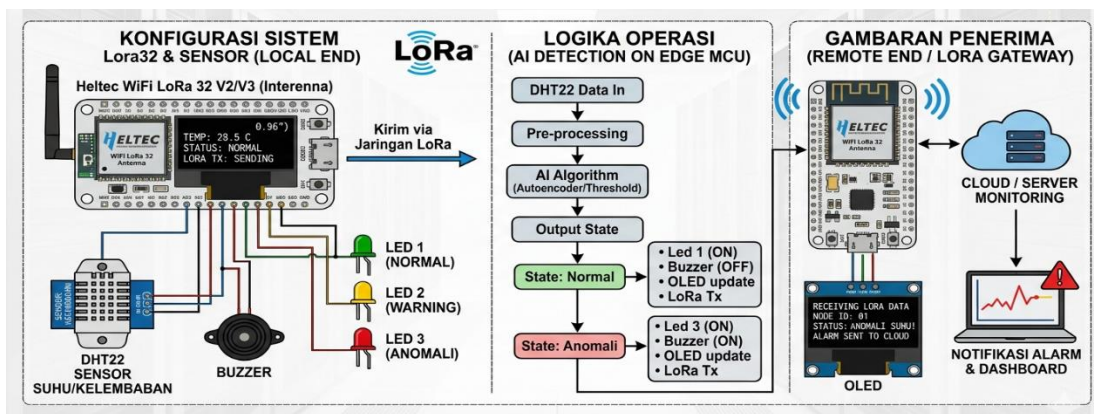


Figure 1. Design and Equipment Scheme

## Power Source

The system is powered using a low-voltage DC power supply suitable for the TTGO LoRa32 device and sensor module. The sensor node and gateway operate using a 5V DC input, which can be supplied through USB power or an external regulated power source. The system is designed to support low-power operation to ensure continuous data collection and transmission. This makes it suitable for long-term deployment in server room monitoring environments without requiring high energy consumption.

## Specifications and Operation of the Device

The system consists of hardware and software components designed to support IoT-based environmental monitoring in server rooms. The hardware specification includes a TTGO LoRa32 microcontroller as both sensor node and gateway device, a DHT22 sensor for measuring temperature and humidity, and a WiFi network module for data transmission from gateway to server. The system operates on low-power LoRa communication to ensure long-range data transmission efficiency. On the software side, the system uses a server application developed with Python (FastAPI) for data handling and processing, a MySQL database for storing sensor data, and an Artificial Intelligence (AI) module for anomaly detection based on temperature and humidity inputs. A web-based dashboard is used for real-time data visualization and monitoring. The operational flow of the system begins with the sensor node periodically reading temperature and humidity data from the environment. The collected data is transmitted via LoRa to the gateway. The gateway then forwards the data to the server using MQTT over WiFi. The server processes and stores the data, then applies AI-based analysis to detect anomalies. Finally, the processed results are displayed on the monitoring dashboard in real time.

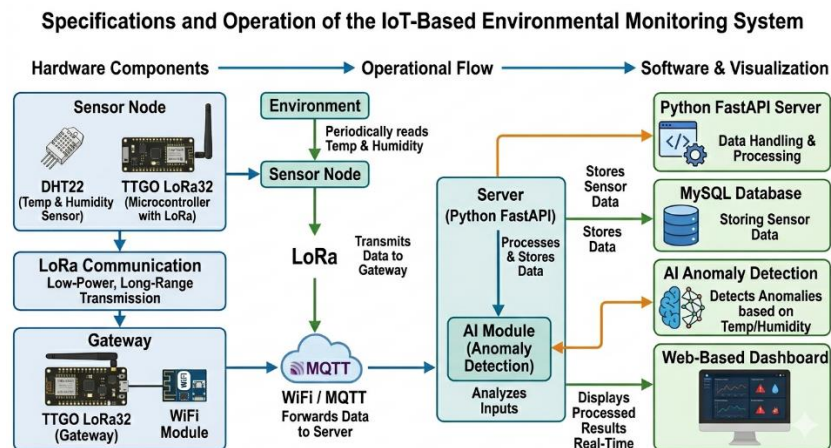


Figure 2. Specifications and Operation of the IoT-Based Environmental Monitoring System

## CHAPTER 5: IMPLEMENTATION

The implementation of the SUSEMON (Temperature and Humidity Server Monitoring) system begins with preparing and integrating the main components required to build an IoT-based monitoring system. The system consists of a sensor node, a LoRa gateway, and a monitoring server that work together to collect, transmit, process, and display server room environmental data. The sensor node is implemented using a TTGO LoRa32 device integrated with a DHT22 sensor, which is responsible for measuring temperature and humidity conditions periodically. The collected data is then transmitted wirelessly using LoRa communication to the gateway device.

The gateway functions as an intermediary device that receives temperature and humidity data from the sensor node through LoRa communication. After receiving the data, the gateway forwards the information to the monitoring server using the MQTT protocol through a WiFi connection. This communication mechanism enables efficient and low-power data transmission while maintaining reliable connectivity between the IoT devices and the server. The monitoring server then processes the incoming data by storing and analyzing the temperature and humidity values to identify abnormal environmental patterns.

Furthermore, the implementation includes an Artificial Intelligence (AI)-based analysis process to detect anomalies in server room conditions. The system applies data processing techniques using temperature and humidity parameters as input to determine the environmental condition status. The analysis results classify the server room condition into three categories: normal, warning, and danger, based on the detected data patterns rather than relying only on fixed threshold values. Methods such as Moving Average and Z-score are applied to support the anomaly detection process and provide early identification of potential environmental problems.

Through this implementation, users can monitor real-time temperature and humidity conditions in the server room through the monitoring dashboard. The system provides continuous environmental observation, helps identify abnormal changes earlier, and reduces the risk of server performance degradation caused by unstable conditions. Therefore, the implementation of SUSEMON combines IoT devices, wireless communication, data processing, and AI-based analysis to create an effective monitoring solution that supports safer and more reliable server room management.

Hardware Implementation

Software Implementation

Database Implementation

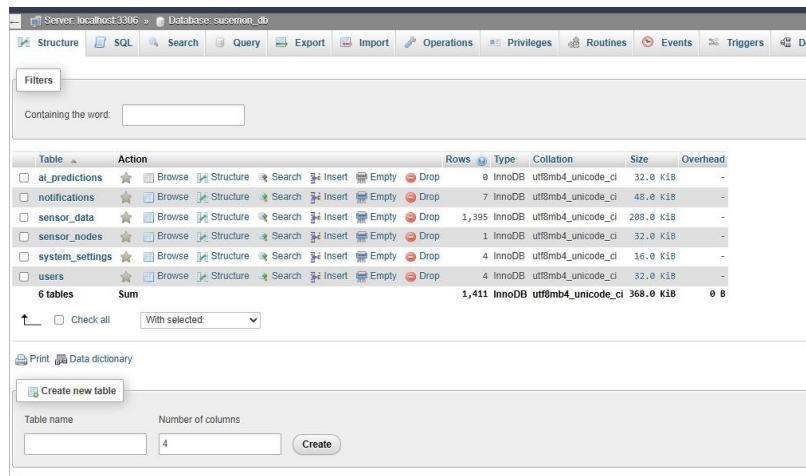


Figure 6. Database SUSEMON

CHAPTER 6: RESULTS & CONCLUSION

The implementation of the sea weather monitoring system incorporating IoT technology and AI has yielded significant outcomes, providing users with accurate real-time sea weather data, including temperature, humidity, air pressure, and altitude. This system has notably enhanced safety at sea by enabling effective monitoring of sea conditions and facilitating swift responses to hazardous situations, consequently reducing the risk of accidents.

Moreover, users can make informed decisions regarding maritime activities based on comprehensive sea weather information and trend analysis, thereby improving operational efficiency. The integration of sustainable energy sources, such as solar panels, ensures continuous system operation while contributing to environmental sustainability. With its intuitive user interface, the system offers a user-friendly experience, further enhancing usability and adoption. In conclusion, the sea weather monitoring system has proven to be a valuable tool for enhancing maritime safety, productivity, and sustainability.

APPENDIX 1: LOGBOOK

Tangkap Layar dari logbook dari siap-PBL dan Gantt Chart



| ID | Tahapan     | Detail Pengerjaan                                                                                                                   | Output                          | Mulai      | Selesai    | Progress | #        |
|----|-------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------|------------|----------|----------|
| 1  | Planning    | Disuksi pertama kali bersama manajer proyek terkait project, menentukan jadwal bimbingan, mencari referensi, memulai pengerjaan RPP | RPP, gambaran kasar aplikasi    | 2026-02-23 | 2026-03-01 | 10%      | 8 Hapori |
| 2  | Planning    | Melanjutkan pengerjaan dokumen RPP, membagi tugas dan role per anggota tim dengan pembuatan gantt chart                             | Dokumen RPP, gantt chart        | 2026-03-02 | 2026-03-08 | 20%      | 8 Hapori |
| 3  | Analysis    | Melakukan pencarian referensi AI (Isolation Forest), flow system, dan alur komunikasi LoRa                                          | Referensi AI, flowchart system  | 2026-03-09 | 2026-03-14 | 25%      | 8 Hapori |
| 4  | Design      | Perancangan arsitektur database (ERD) dan pembuatan wireframe low-fidelity dashboard mobile                                         | ERD MySQL, wireframe UI         | 2026-03-30 | 2026-04-04 | 30%      | 8 Hapori |
| 5  | Design      | Pembuatan desain high-fidelity di Figma, penyajian komponen hardware (ESP32, SX1276, DHT22)                                         | Mockup Figma, komponen hardware | 2026-04-06 | 2026-04-11 | 40%      | 8 Hapori |
| 6  | Development | Pengembangan firmware node (LoRa transmitter) dan gateway (LoRa-to-MQTT bridge)                                                     | Firmware jns, koneksi LoRa      | 2026-04-27 | 2026-05-03 | 55%      | 8 Hapori |
| 7  | Development | Implementasi UI ke Flutter widget dan setup awal backend FastAPI dengan koneksi database                                            | Flutter layout, FastAPI base    | 2026-05-04 | 2026-05-10 | 65%      | 8 Hapori |
| 8  | Development | Pengembangan MQTT subscriber, riset mendalam Isolation Forest, dan layouting grafik fl_chart                                        | MQTT listener, script AI alpha  | 2026-05-11 | 2026-05-17 | 75%      | 8 Hapori |
| 9  | Development | Integrasi REST API ke Flutter, modul Data Cleaning AI, dan optimasi query database history                                          | API service, preprocessing AI   | 2026-05-18 | 2026-05-24 | 85%      | 8 Hapori |
| 10 | Development | Integrasi AI ke pipeline real-time, implementasi WebSocket, dan finalisasi casing hardware                                          | AI real-time, WebSocket UI      | 2026-05-25 | 2026-05-31 | 95%      | 8 Hapori |

| ID | Tahapan | Detail Pengerjaan                                                                       | Output                         | Mulai      | Selesai    | Progress | #        |
|----|---------|-----------------------------------------------------------------------------------------|--------------------------------|------------|------------|----------|----------|
| 11 | Testing | Pengumpulan dataset suhu, pelabalan data, dan pelatihan model AI Isolation Forest       | Dataset final, model AI gkl    | 2026-06-01 | 2026-06-07 | 96%      | 8 Hapori |
| 12 | Testing | Uji akurasi sensor DHT22 dengan kalibrator dan range test (indoor/outdoor)              | Laporan kalibrasi & range test | 2026-06-08 | 2026-06-14 | 97%      | 8 Hapori |
| 13 | Review  | Stress test API backend, usability testing dashboard mobile, dan instalasi fisik sensor | Bug report, sistem terpasang   | 2026-06-15 | 2026-06-21 | 98%      | 8 Hapori |

(TINGGAL GANTT CHART)

## APPENDIX 2: PBL WORK DOCUMENTATION

Foto-foto proses dan aktivitas pengerjaan PBL





